

Quick Check

Lesson 1: Meeting the Phenomenon: Mr. Wafula's Maize and Beans

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 2.2: Growth and Development in Plants | Lesson 1

On Mr. Wafula's farm near Kitale, 100 bean seeds were planted. By Day 14, 80 of them had germinated. How many bean seeds had NOT germinated by Day 14?

A 8

B 80

C 20

D 12

In the time-lapse records, the bean radicle appeared on Day 3 and the bean shoot came above the soil on Day 8. The maize radicle appeared on Day 4 and the maize shoot came above the soil on Day 7. Which statement matches these records?

- A** Maize was first for both the radicle and the shoot above soil
- B** Both crops had their radicle and shoot above soil on the same days
- C** The bean radicle appeared first, but the maize shoot came above the soil first
- D** Bean was first for both the radicle and the shoot above soil

A group is studying the bean time-lapse. Their table shows the radicle appeared on Day 3 and the shoot came above the soil on Day 8. What can they say about the bean seed on Day 5?

- A** It had not germinated yet because no shoot was above the soil
- B** It had failed to germinate because nothing was visible above the soil
- C** It had both germinated and emerged
- D** It had germinated because the radicle had appeared, but it had not yet emerged above the soil

Learners watched a bean plant for 25 days. Which observation shows that plant growth is more than just getting taller?

- A** Maize was taller than beans from Day 14 onward
- B** The stem became thicker and the bean plant developed branches by Day 25
- C** More maize seeds than bean seeds germinated by Day 14
- D** The bean radicle appeared on Day 3

Mr. Wafula planted maize and beans on the same day, 3 cm deep, after 25 mm of rain, in the same field. Some healthy-looking seeds did not germinate. At the end of this first lesson, which statement is the most honest one?

- A** Healthy-looking seeds that fail must have been damaged by pests
- B** The seeds failed because they were planted too deep
- C** We do not yet know the reason; it is a question we need to investigate
- D** The seeds failed because beans need more water than maize

Groups are writing questions for the Driving Question Board about Mr. Wafula's maize and beans. Which question is written in a way that could be investigated?

A Is farming a hard job?

B Why is maize better than beans?

C Who planted the seeds first?

D Does the amount of water given to bean seeds change how many of them germinate by Day 14?

For an exit ticket, a learner must write an observation, not an explanation, about Mr. Wafula's crops. Which statement is an observation?

- A** Some seeds failed because the bean seed has less food inside
- B** Maize emerged above the soil on Day 7 and beans on Day 8
- C** Maize emerged first because it is stronger than beans
- D** Beans need more water to germinate

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).